

- Power requirement of 2 HDDs =  $2 \times 10 \text{ W}$  (approx.) = 20 W
- Power requirement of a GTX 1080 Ti = 260 W

Add them up and you get 371 W. If you were to use a Power Supply unit with a 500W (or VA) rating, you're left with 129W, right? Wrong! That 500W you have would operate at about 80-90% efficiency. Let's assume it's working at ~85% efficiency. This means the power it can provide to your components comes to  $(500\text{W} \times 85/100) = 425\text{W}$ . So in a practical world, your SMPS has only around 55 watts of energy to spare. If you were to add one more component which happens to use that much energy, you will have pushed your SMPS to its limit. While it feels good to utilise your components to the max, you don't want to do that with a power supply unit.

### Key points of a power supply unit

Power supply units (PSU) are central to a computer's operation. Right from when you hit the power-on button and all the components start up. The notable power draw amongst them are the devices which utilize motors – hard disk drives and all the fans mounted at various places (cabinet, CPU cooler, graphics card heatsink etc.). Power drawn from the power supply unit in the first second can go up to twice the power rating of all devices. At the same time, voltage fluctuations are pretty much a norm in India. A good power supply unit takes care of this and prevents itself and other components from failing and getting damaged.

We must stress on the importance of a good PSU – if the PSU fails or gets damaged, it can potentially damage any (all) components of a PC. All devices have their own voltage and power ratings. Extra voltage or current on any of the lines can fry up a component while a drop in either will either shut down the device or put them in an irrecoverable state for that session. It's the PSU's job to ensure that these things do not happen.

PSUs are marked with their operational voltage ratings, total power output, efficiency and number of cables and ports provided. A PSU is supposed to have:

1. 24 (20+4) pin connector – they go into the power input socket on the motherboard.
2. CPU (4+4) pin connector – they supply power to the CPU and go into the input socket normally located close to the CPU socket on the board.
3. PCI-e (6+2) pin connector – these power graphics cards and go into them.
4. SATA power connectors – these go into your hard disks and SATA based Optical Drives.